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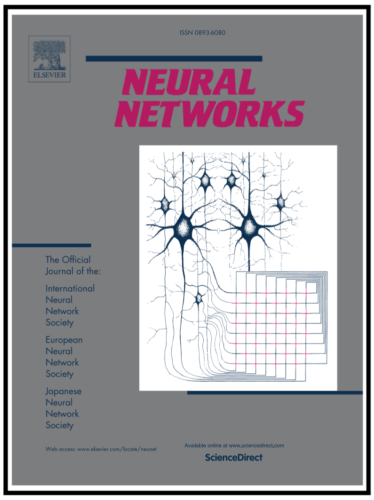

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An Efficient and Accurate YOLO-Based Framework for Air-to-Air UAV Detection

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Abstract

Robust Air-to-Air (A2A) detection of unmanned aerial vehicles (UAVs) is critical for the safety of low-altitude airspace. This paper proposes a lightweight yet highly accurate YOLO-based detector to address this challenge. Our framework integrates three novel modules: the ASF-P for enhanced multi-scale feature fusion, the AB-CGLU for attention-boosted feature representation, and the ADown for efficient downsampling. This design effectively tackles small UAV detection in complex backgrounds while maintaining a compact architecture. On the primary dataset, our model achieves state-of-the-art performance with 98.1% precision, 84.4% recall, 91.6% mAP@0.5, and 57.1% mAP@0.5:0.95, while requiring only 2.03M parameters and 4.7MB of storage. It outperforms YOLOv8n by a large margin (e.g., +12.6% mAP@0.5) and also surpasses heavier YOLO variants in accuracy. Critically, cross-dataset validation over three independent runs on challenging unseen benchmarks confirms its strong generalization capability. This work provides an efficient and reliable solution for real-time A2A UAV perception, with immediate potential for deployment on resource-constrained autonomous aerial platforms.

Keywords: Air-to-Air UAV Detection; Small Object Detection; Lightweight Model; YOLO; Multi-Scale Fusion; Attention Mechanism

1. Introduction

In recent years, the rapid deployment of Unmanned Aerial Vehicles (UAVs) across civil applications, including logistics, environmental monitoring, agricultural inspection, urban security, and aerial traffic management, has led to increasingly congested and heterogeneous low-altitude airspace environments [1]. In such scenarios, UAVs must operate safely while perceiving surrounding dynamic airborne targets in real time, particularly in missions involving multi-UAV coordination, urban air mobility (UAM), and autonomous obstacle avoidance [2]. This challenge becomes more pronounced in low-altitude flight conditions with complex environmental structures, where small airborne UAV targets are difficult to detect yet critical for reliable navigation and collision avoidance [3].

According to observation perspectives and mission requirements, UAV detection research can be broadly categorized into Ground-to-Air (G2A) and Air-to-Air (A2A) paradigms [4]. Among them, G2A detection is the more mature research direction [5]. It relies on ground-based sensors-such as fixed cameras, radar, or

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multispectral devices-to detect airborne UAV targets from stable viewpoints with wide observation coverage. This stable imaging configuration makes G2A methods highly effective for applications including low-altitude surveillance, airport clearance management, and urban airspace regulation [6]. Benefiting from the rapid advancement of computer vision and deep learning, image-based G2A approaches have achieved high accuracy and robustness on large-scale public benchmarks such as Anti-UAV [7].

In contrast to the relatively stable observation conditions of G2A, Air-to-Air (A2A) UAV detection refers to the task in which an airborne UAV serves as the observation platform to perceive and identify other UAV targets in dynamic aerial environments. The continuous motion of the onboard camera leads to constantly changing viewpoints, complex backgrounds, motion blur, and severe scale variations, creating a highly challenging robustness evaluation setting [8]. As a result, targets often appear as small, weak objects at long distances [9] and are prone to background fusion and occlusion. These characteristics make A2A detection fundamentally more challenging than conventional G2A scenarios and impose strict requirements on robust and low-latency small object detection.

As a key component of autonomous perception in intelligent unmanned systems, A2A UAV detection has recently attracted increasing attention at the intersection of computer vision and flight control [10]. To address this challenging task, the methodological evolution has transitioned from traditional vision to deep learning paradigms [11]. Early studies mainly relied on traditional image processing and feature matching methods. For example, Zhou et al. proposed a UAV recognition method based on background subtraction and morphological operations, which performed adequately in simple static scenes but struggled in complex environments [12]. Wang et al. utilized HOG features combined with an SVM classifier for A2A target detection. However, this approach was highly sensitive to target rotation and scale variations [13]. With the integration of deep learning, significant progress has been achieved in UAV detection through advances in network architecture design, detection accuracy, and adaptation to edge deployment [14]. Improvements based on the You Only Look Once (YOLO) family of models have become mainstream due to their favourable balance between accuracy and real-time performance. For instance, Drone-YOLO integrates multi-layer PAFPN and small-object detection heads, achieving a substantial improvement over YOLOv5 on the VisDrone dataset [15]. Meanwhile, Transformer-based architectures have also been introduced into aerial vision tasks to enhance global contextual modelling [16-18].

Despite these advancements, most existing methods are primarily designed and validated under G2A observation settings or general aerial imagery, where imaging viewpoints are relatively stable and background interference is limited. These assumptions do not hold in A2A scenarios. As a result, even state-of-the-art YOLO and Transformer-based detectors experience noticeable performance degradation when directly applied to A2A UAV detection tasks. Furthermore, their computational cost often contradicts the stringent efficiency requirements of onboard processing [19].

The challenges of A2A UAV detection can be summarized into three fundamental aspects. First, extreme small-object characteristics: target UAVs often occupy only a few pixels due to long observation distances, resulting in weak textures and blurred boundaries that are easily overwhelmed by background noise [20-22]. Second, severe background interference: complex background such as sky, mountains, horizons, and urban structures, exhibit similar textures and low contrast, leading to substantial overlap between foreground and background feature distributions [23]. Third, strict real-time and deployment constraints: A2A detection systems are typically deployed on resource-limited airborne platforms, where computational efficiency, lightweight design, and low-latency inference are mandatory [24-26].

To address these issues, this study proposes an efficient and accurate UAV detection framework specifically tailored for complex A2A scenarios. The proposed architecture introduces targeted enhancements to the backbone, neck, and downsampling stages of a YOLO-based detector to improve contextual modeling, scale-

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aware feature fusion, and computational efficiency. The main contributions of this work are summarized as follows:

1. **Lightweight yet robust detection framework for A2A UAVs:** We propose an efficient YOLO-based architecture that enhances contextual modeling and multi-source feature integration, enabling accurate detection of small UAV targets under cluttered backgrounds while maintaining low computational cost. This framework directly addresses the challenge of detecting weak and distant objects in dynamic aerial environments.
2. **Attention-boosted cross-gated linear unit module (AB-CGLU) for small-object representation:** To enhance feature discrimination for tiny UAV targets, AB-CGLU integrates convolutional encoding with additive self-attention, allowing adaptive channel selection based on local context. This module strengthens weak visual cues and improves small-object detection accuracy, directly mitigating the extreme small-object challenge.
3. **Attention-guided scale-aware feature fusion (ASF-P) for multi-scale responsiveness:** ASF-P combines scale-sequence feature fusion, triple feature encoding, and channel-position attention to capture scale-variant target features effectively. By fusing information across multiple scales and emphasizing relevant channels, ASF-P reduces false positives and improves detection robustness against severe background interference.
4. **Adaptive downsampling module (ADown) for efficient inference on resource-limited platforms:** ADown strategically compresses spatial dimensions while preserving informative features, reducing computational overhead and enabling low-latency inference. This design directly addresses the real-time and deployment constraints inherent in A2A UAV detection systems.

The remainder of this paper is organized as follows: Section 2 presents the overall architecture of the proposed model, detailing the design and integration of each core module. Section 3 describes the experimental settings, dataset details, and evaluation metrics, and provides comprehensive analyses through ablation studies, comparative experiments, and visualization results. Section 4 discusses and summarizes the findings of this study. Finally, Section 5 concludes the work and outlines future research directions.

2. Methodology

2.1. Overview of the YOLOv8 Architecture

YOLOv8, introduced by Ultralytics in 2023 [27], marks a significant evolution in the YOLO series for real-time object detection. It retains the proven three-stage architecture-Backbone, Neck, and Head-while introducing key refinements for improved performance and efficiency. The Backbone is streamlined convolutional neural network that extracts hierarchical features. The Neck fuses these multi-scale feature representations to enhance detection across varying object sizes. Finally, the Head performs classification and bounding box regression to localize and identify targets. Notably, YOLOv8 replaces Cross Stage Partial (CSP) modules with a lighter, more scalable design and employs advanced convolution and normalization strategies, achieving a better speed-accuracy trade-off than its predecessors.

2.2. Overall Architecture of Proposed Framework

To address the inherent challenges in A2A UAV detection-such as small object sizes, significant scale variations, and complex backgrounds, this study proposes an efficient and accurate detection architecture based on YOLOv8n. The overall architecture is illustrated in Fig 1.

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The proposed model retains the three-stage structure of YOLOv8, consisting of a Backbone, a Neck, and a Head, while introducing three targeted enhancements specifically designed for A2A scenarios:

(1) In the Backbone, we enhance contextual feature interaction to improve the representation capability for small and distant targets (implemented through the proposed AB-CGLU module).

(2) In the Neck, we redesign the feature pyramid structure to achieve more robust and scale-aware multi-scale feature fusion (implemented through the proposed ASF-P module).

(3) In the downsampling process, we replace standard strided convolutions with an adaptive strategy to preserve critical information while reducing computational complexity (implemented through the proposed ADown module).

These targeted improvements enable the framework to maintain the efficiency of YOLO-based detectors while significantly enhancing detection accuracy and robustness for real-time A2A UAV detection tasks. The detailed designs of these three modules are presented in Sections 2.3, 2.4, and 2.5, respectively.

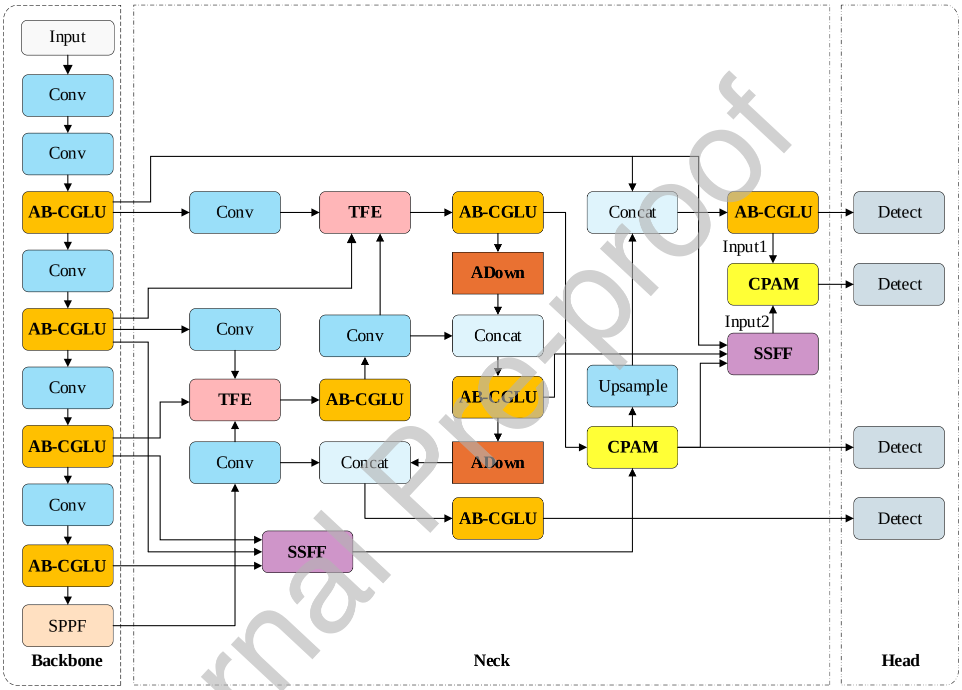

Fig 1 Overall architecture of proposed framework.

2.3. Design of the AB-CGLU Module

The original YOLOv8 backbone has limitations for A2A detection. First, its LSKAttention module enhances spatial modeling but lacks effective channel-wise interaction. It also incurs high computational costs, hindering deployment. Second, the C2f module relies on convolution for local feature extraction. This makes it struggle to model the long-range dependencies that are crucial for detecting small objects in complex backgrounds [28].

To overcome these issues, we propose the Attention-Boosted Cross-Gated Linear Unit (AB-CGLU) module to replace the original C2f blocks in the YOLOv8n backbone [29]. Built upon the AdditiveBlock structure, the AB-CGLU module retains dual-domain (spatial and channel) interaction while introducing a novel Cross-Gated Linear Unit (CGLU) in place of the standard Multi-Layer Perceptron (MLP). This key substitution enhances selective feature representation and improves adaptability to lightweight A2A detection tasks.

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The AB-CGLU module comprises three components: Integration, Convolutional Additive Token Mixer (CATM) and CGLU. Fig. 2 illustrates its overall structure, with details in the following subsections.

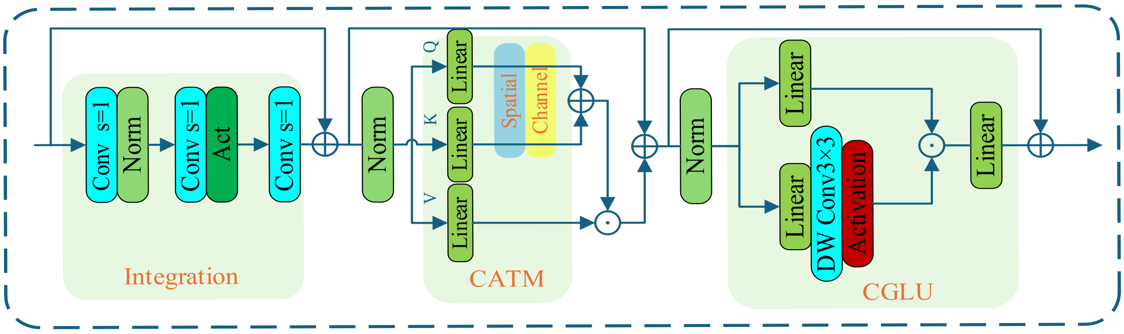

Fig 2. Structure of AB-CGLU module.

2.3.1 Integration

This component employs three stacked layers of 3×3 depthwise separable convolutions, each followed by ReLU activation. The stacked design progressively expands the receptive field, enabling the network to integrate multi-scale local features effectively. This process enhances spatial contextual modeling of target regions, and provides a high-quality intermediate feature representation for subsequent modules.

2.3.2 Convolutional Additive Token Mixer (CATM)

The CATM module enables efficient dual-domain interaction by integrating complementary attention mechanisms from both spatial and channel domains.

- **Spatial Branch:** This branch employs a 3×3 convolution followed by a 1×1 convolution to extract rich local contextual features. Batch Normalization (BN) and ReLU activation are applied after each convolution to stabilize training and enhance nonlinear representation.
- **Channel Branch:** Inspired by SENet [30], this branch utilizes global average pooling to compress spatial information and generate channel-wise statistics, which serve as global descriptors.

The resulting spatial and channel attention maps are then fused through an additive operation, producing a unified context-aware feature representation. This fusion process, governed by an additive similarity function and the final output of CATM are defined as follows:

$$\Phi(x) = x_s + x_c \tag{1}$$

$$O = \Gamma(\Phi(x)) \odot V \tag{2}$$

Where, x_s represents spatial features, x_c represents the channel features. Γ denotes a linear transformation, $\odot$ is the element-wise multiplication, and V is the value-branch feature map.

2.3.3 Convolutional Gated Linear Unit (CGLU)

The original AdditiveBlock (AB) employs a standard Multi-Layer Perceptron (MLP) as its third component (Fig. 3). While effective for nonlinear transformation, the MLP has inherent limitations for our task: it operates independently across channels, failing to model critical inter-channel dependencies; its transformation is static, lacking input-adaptive modulation; and its dense connections incur high parameter and computational costs, especially for high-dimensional features, which burdens training and inference efficiency.

To overcome these limitations, we introduce CGLU to replace the MLP within the AB. The CGLU is designed to generate more discriminative feature representations, better capture UAV-specific characteristics, and thereby improve detection accuracy and robustness, particularly in cluttered backgrounds [31]. The core of

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CGLU is a learned, input-dependent gating mechanism that dynamically regulates information flow across both spatial and channel dimensions. A gating vector, generated via convolution, performs element-wise multiplication with the input features, selectively amplifying informative patterns while suppressing less relevant ones. This adaptive process is formulated as:

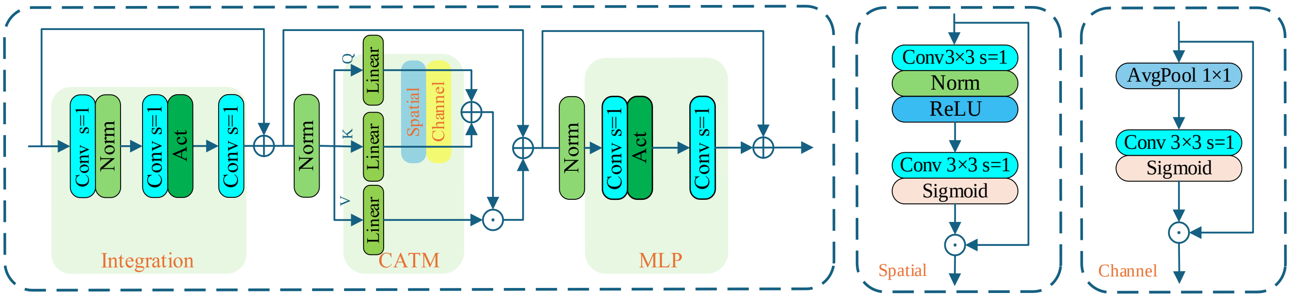

Fig 3. Architecture of the original AdditiveBlock (AB).

$$\mathrm{CGLU}(X) = \sigma(W_g * X + b_g) \odot (W_f * X + b_f) \tag{3}$$

Where, X is the input feature map, W_g, W_f and b_g, b_f are the convolutional weights and biases for the gating and feature branches, respectively, * denotes convolution, ⊙ denotes element-wise multiplication, and σ (·) is a nonlinear activation function (e.g., Sigmoid).

By integrating the CGLU into the AB-CGLU module (replacing the original C2f block), the model gains a significantly enhanced ability to capture subtle features of small objects and maintains superior discriminative robustness in complex scenes, all within a lightweight architecture conducive to efficient deployment.

2.4. Design of ASF-P Module

The neck structure of YOLOv8, which integrates concepts from the Feature Pyramid Network (FPN) and Path Aggregation Network (PANet), still exhibits limitations in multi-scale feature modeling and fusion. This is particularly evident in its weak representation of small objects against complex backgrounds, which constrains detection accuracy in UAV-centric tasks dominated by small targets [32].

To address this issue, we propose a novel attention-enhanced, scale-aware feature fusion module called ASF-P. This module introduces a dedicated detection head called P2. By incorporating P2, which is specifically designed for small objects, and by integrating multiple attention-enhancement modules, the ASF-P module significantly improves both the capacity for feature representation and detection performance across multiple scales. The ASF-P module comprises four key components: the Scale Sequence Feature Fusion (SSFF) module, the Triple Feature Encoding (TFE) module, the Channel and Positional Attention (CPAM) module, and an additional dedicated detection head (P2). The structure and functionality of each component are described as follows.

2.4.1 Scale Sequence Feature Fusion (SSFF) module

As illustrated in Fig 4, the SSFF module first takes three multi-scales feature maps (P3–P5) from the backbone. Each map passes through a 1×1 convolutions to unify channel dimensions and reduce redundancy. All feature maps are then upsampled to the resolution of the P2 layer via nearest-neighbor interpolation, which preserves sharp edges for small objects better than bilinear interpolation.

$$F_\sigma(i,j) = \sum\nolimits_{u} \sum\nolimits_{v} (i-u, j-v) \times G_\sigma(u,v) \tag{4}$$

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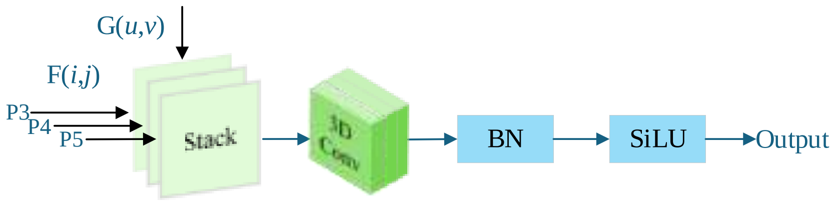

Fig 4. Structure of SSFF module.

Where, f denotes the original 2D feature map, and F_σ represents the smoothed result. The Gaussian kernel $G_\sigma(u,v)$ is defined as:

$$G_\sigma(u,v) = \frac{1}{2\pi\sigma^2} e^{-(u^2+v^2)/2\sigma^2} \tag{5}$$

The three smoothed feature levels are expanded into a 4D tensor and concatenated along the depth dimension. A 3D convolution then models inter-scale dependencies across this unified representation. Finally, 3D batch normalization and the SiLU activation function refine the fused features to enhance nonlinear expressiveness. By effectively integrating shallow spatial details with deep semantic context, the SSFF module significantly improvs the model’s sensitivity to objects across scales.

2.4.2 Triple Feature Encoding (TFE) module

The TFE module enhances the model’s ability to detect multi-scale UAV targets in complex scenes by efficiently fusing multi-scale features and hierarchically encoding them. As illustrated in Fig 5 (where C denotes the number of channels and S the spatial size of the feature map), the TFE module processes large-, medium-, and small-scale feature maps separately to achieve hierarchical feature fusion across different scales.

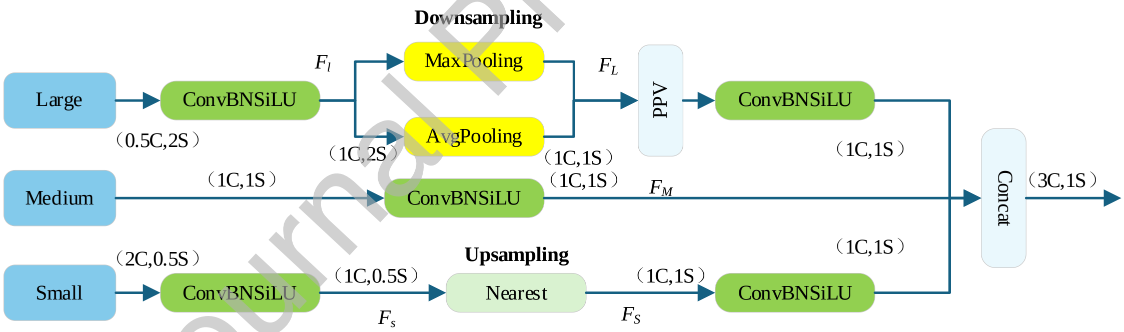

Fig 5. Structure of TFE module.

For the large-scale feature map F_l, salient regional information is extracted using a hybrid downsampling strategy that combines Max Pooling and Average Pooling:

$$F_L = \mathrm{Conv}(\mathrm{MaxPool}(F_l) + \mathrm{AvgPool}(F_l)) \tag{6}$$

For the small-scale feature map F_s, nearest-neighbor interpolation is used to upsample the feature maps to a unified resolution, avoiding the detail loss often caused by bilinear method:

$$F_S(i,j) = F_s\left(\left\lfloor \frac{i}{s} \right\rfloor, \left\lfloor \frac{j}{s} \right\rfloor\right) \tag{7}$$

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Where, s is the scaling factor, and $\lfloor\cdot\rfloor$ denotes the floor operation. This preserves local pixel integrity for small targets.

Finally, the processed features from all three scales are concatenated along the channel dimension and fused via convolution:

$$F_{\mathrm{TFE}} = \mathrm{Conv}(\mathrm{Concat}(F_L, F_M, F_S)) \tag{8}$$

By enabling effective cross-scale interaction, the TFE module mitigates missed detections caused by scale variation. When combined with the EIOU loss function, which offers fine-grained optimization for aspect ratio variations, the TFE module significantly enhances the model's robustness and localization accuracy for objects across scales, particularly in cluttered or dynamic backgrounds.

2.4.3 Channel and Positional Attention (CPAM) Module

The structure of CPAM is illustrated in Fig 6. This module consists of two key components: channel attention and position attention. The channel attention enhances feature channels strongly correlated with target categories, thereby boosting classification confidence. In parallel, the position attention suppresses background noise and sharpens spatial focus, improving localization accuracy. The dual-attention branch facilitates the enhanced representation of multi-scale features.

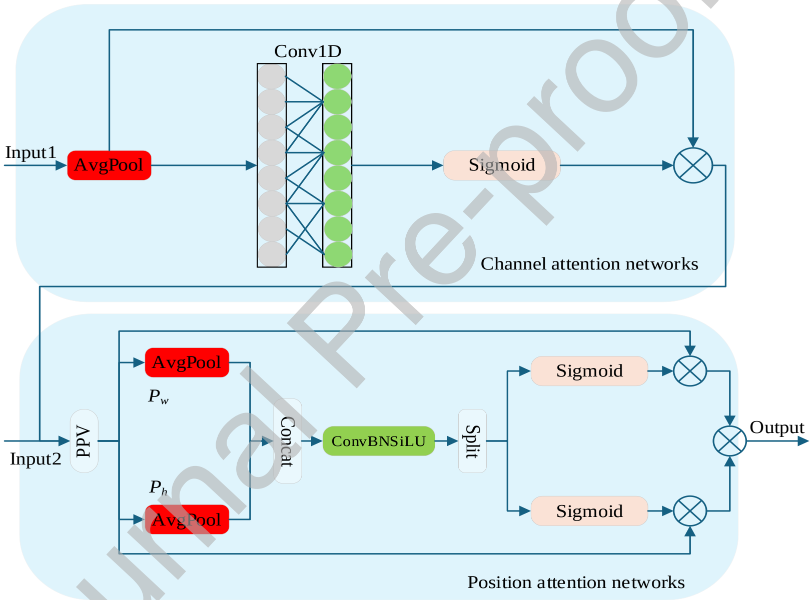

Fig 6. Structure of CPAM module: where, w and h denote the width and height of the feature map, and $\otimes$ represents the Hadamard (element-wise) product.

(1) Channel attention: This branch models cross-channel dependencies via 1D convolution without downsampling. The convolution kernel size is adaptively determined by the channel dimension C:

$$k = \Psi(C) = \left\lfloor \frac{\log_2(C) + b}{\gamma} \right\rfloor_{\mathrm{odd}} \tag{9}$$

where $\lfloor\cdot\rfloor_{\mathrm{odd}}$ denotes rounding to the nearest odd integer, the hyperparameters γ=2 and b=1 control the ratio between the number of channels and the convolution kernel size. This design allows higher-channel feature maps to capture global dependencies, while lower-channel features focus on local patterns, balancing efficiency and representation.

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Let the enhanced channel-attentive feature map be:

$$E = F_{\mathrm{SSFF}} + \mathrm{F}_{\mathrm{TFE}}^{\mathrm{att}} \tag{10}$$

(2) Position attention: This branch captures global spatial context. First, global average pooling is applied along the horizontal and vertical axes separately:

$$p_w(i) = \frac{1}{H} \sum_{0 \le j < H} E(i, j) \tag{11}$$

$$p_h(j) = \frac{1}{W} \sum_{0 \le i < W} E(i, j) \tag{12}$$

Where, W and H represent the width and height of the feature map E, p_w and p_h encode the global contextual distribution along the horizontal and vertical directions, respectively. The two context vectors are then concatenated and passed through a convolutional layer to generate spatial attention weights:

$$P(a_w, a_h) = \mathrm{Conv}(\mathrm{Concat}(p_w, p_h)) \tag{13}$$

Where, $P(a_w,a_h)$ denotes the output of position attention coordinates. The final attention-enhanced output feature map is computed as:

$$F_{CPAM} = E \times s_w \times s_h \tag{14}$$

$$s_w = \mathrm{Split}(a_w) \tag{15}$$

$$s_{\mathrm{h}} = \mathrm{Split}(a_{\mathrm{h}}) \tag{16}$$

Where, s_w and s_h are the spatial attention components derived from the split operation.

In A2A UAV detection, CPAM's channel attention highlights fine-grained, high-frequency features of tiny targets, while its position attention pinpoints the target center. This dual-attention branch effectively mitigates misclassification, reducing both missed detections and misclassification, and ensures the detection head performs operates on highly discriminative features for precise localization and classification.

2.4.4 Small Object Detection Head (P2)

The original YOLOv8 employs three detection heads (P3, P4, and P5) for detect small, medium, and large-scale targets, respectively. However, its performance on small objects is limited due to their weak semantic representation, and the risk of feature dilution during repeated convolution operations. A key reason is the 8× downsampling of the P3 feature map, where small targets occupy too few pixels in the P3 feature map to be reliably detected.

To address this limitation, we introduce an additional small object detection head at the P2 level, which corresponds to a 4× downsampled feature map. This new detection head preserves richer spatial and fine-grained details of small target. When integrated into the ASF-P module, which already provides multi-scale fusion and dual-domain attention enhancement, the P2 head significantly boosts the accuracy and robustness of small target detection.

In summary, the proposed ASF-P module unifies multi-scale feature fusion, channel-spatial attention enhancement, and the new P2 detection head into a cohesive and efficient detection pipeline. This integrated design strengthens small UAV detection in complex scenes, improves feature expressiveness and localization precision, and maintains high computational efficiency for practical deployment.

2.5. ADown Module

Standard strided convolution in the YOLOv8 neck, while efficient for downsampling, tends to lose fine-grained features that are crucial for small target detection. To better preserve multi-scale information during resolution reduction, we adopt the ADown module [33] to replace part of the original downsampling operations.

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ADown combines average pooling with lightweight convolutional branches to compress spatial dimensions while retaining richer structural details. The structure of ADown module is illustrated in Fig 7.

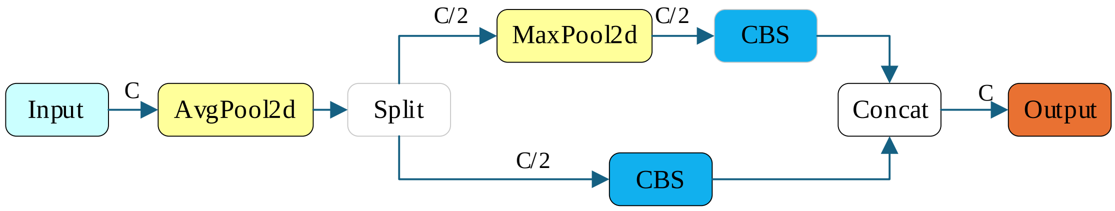

Fig 7. Architecture of the ADown module.

As illustrated in Fig. 7, the input feature map first undergoes a stride-2 average pooling (AvgPool2d) to halve its resolution while preserving global structure. The resulting feature map is then split along the channel dimension into two parallel branches:

(1) The first branch directly employs a lightweight Conv-BN-SiLU (CBS) module to extract local discriminative features.

(2) The second branch applies an additional AvgPool2d before its CBS block, enhancing sensitivity to low-frequency patterns and contour structures.

The outputs of both branches are concatenated along the channel axis to form the final downsampled feature map for subsequent layers.

This hybrid design allows ADown module to maintain high sensitivity to objects of varying structures. Moreover, it is composed entirely of lightweight operators, leading to significantly fewer parameters and lower FLOPs than standard strided convolution. These characteristics makes ADown module particularly suitable for resource-constrained A2A UAV detection systems, where preserving detail and computational efficiency are both paramount.

3. Experiments and Results

3.1. Experimental Setup

(1) Hardware and Software: All experiments were conducted on a workstation with an NVIDIA GeForce RTX 4060 GPU (8GB VRAM), an Intel Core i7-12850HX CPU, and 16 GB RAM. The code was implemented in Python 3.9 with PyTorch 2.0 and CUDA 11.8 on a Windows 11 system.

(2) Training Configuration: We trained the model using the stochastic gradient descent (SGD) optimizer with an initial learning rate of 0.01, momentum of 0.937, and weight decay of 5×10^{-4}. Input images were resized to 640×640, and training was conducted for 200 epochs with a batch size of 8. The learning rate was dynamically adjusted using a cosine annealing schedule.

(3) Augmentation and Inference: To ensure fair comparisons, all baseline models were trained and evaluated under the same settings. During training, commonly used data augmentation techniques (including Mosaic, MixUp, random horizontal flipping, and color jittering) were applied to improve generalization. During inference, non-maximum suppression (NMS) was used to eliminate redundant predictions, and a uniform confidence threshold of 0.25 was adopted across all models.

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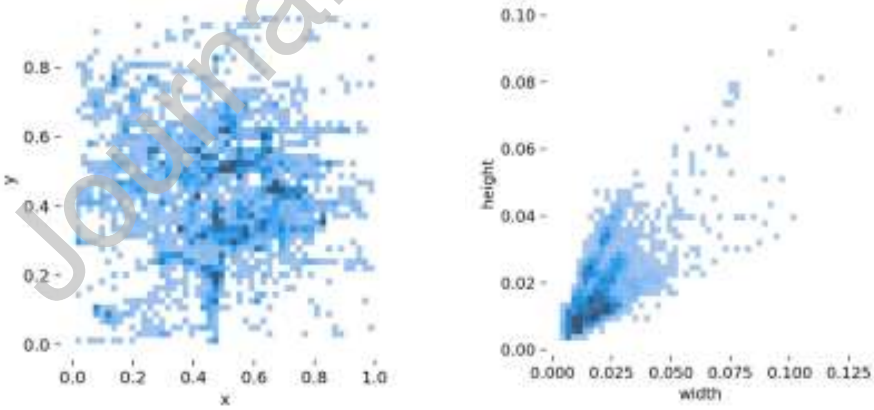
3.2. Dataset Details

We employ the publicly available DeTFLy dataset [34], a large-scale benchmark specifically designed for A2A UAV detection. It contains 13,271 high-resolution images (3840 × 2160 pixels) captured using a DJI Mavic UAV, each image annotated with bounding boxes. The dataset encompasses diverse backgrounds (Sky, Urban, Field, and Mountain, each ~ 20-30%), viewing angles (frontal, top, and bottom) and lighting conditions spanning from morning to evening. It includes challenging cases: 10.8% with strong/weak lighting, 11.2% with motion blur, and 0.8% with partial occlusion, making it suitable for robustness evaluation. Prior work achieved a maximum AP of 90.1% on this dataset, underscoring its complexity. For our experiments, the dataset is divided into training, validating and testing sets with an 8:1:1 ratio. Fig 8 presents image samples from the four primary background scenarios.

(a) (b) (c) (d)

Fig 8. Sample images from dataset: (a) Mountain, (b) Urban, (c) Field, (d)Sky

Fig 9 visualizes the annotation statistics. Fig 9(a) shows the normalized center coordinates of all bounding boxes. Targets are distributed across the image but concentrate markedly in the central region, consistent with typical framing in aerial photography. Fig. 9(b) shows the normalized width and height distribution. Most targets are small, with widths between 0.01-0.05 and heights between 0.01-0.04 of the image dimensions. The diagonal trend indicates relatively uniform aspect ratios, reflecting the stable shape of most UAVs. While dominated by small objects, the dataset includes a range of scales, providing valuable variation for training and evaluation.



(a) (b)

Fig 9. Annotation statistics of the DeTFLy dataset: (a) Distribution of bounding box centers. (b) Distribution of bounding box widths and heights.

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In summary, the DeTFLy dataset is characterized by predominantly small target sizes, a center-dense spatial distribution, and diverse scale variations. These characteristics present significant challenges for detection models, particularly in multi-scale feature learning and precise localization.

3.3. Evaluation Metrics

The proposed model is evaluated from two key perspectives: detection accuracy and model lightweighting.

Detection accuracy is assessed using Precision (P), Recall (R), Average Precision (AP), and two widely adopted mean Average Precision metrics: mAP@0.5 and mAP@0.5:0.95. Specifically, mAP@0.5 is the mean AP across all categories at an Intersection over Union (IoU) threshold of 0.5, reflecting performance under a lenient matching criterion. In contrast, mAP@0.5:0.95 is obtained by averaging the AP over 10 IoU thresholds ranging from 0.5 to 0.95 (step 0.05). This provides a more rigorous assessment of localization robustness and is the mainstream benchmarks in object detection. The mathematical definitions are:

$$\mathrm{P} = \frac{TP}{TP+FP} \tag{17}$$

$$\mathrm{R} = \frac{TP}{TP+FN} \tag{18}$$

$$\mathrm{AP} = \int_0^0 P(R)\, dR \tag{19}$$

where TP refers to the number of true positives (correctly detected UAVs), FP is the number of false positives (non-UAV objects misclassified as UAVs), FN is the number of false negatives (missed UAVs), C is the total number of classes.

Model lightweighting is quantified, considering the computational constraints of aerial platforms, via three metrics:

Number of Parameters (Params): The total number of trainable and storable weights, calculated as:

$$\mathrm{Params} = O\left(\sum_{i=1}^{n} M_i^2 * K_i^2 * C_{i-1} * C_i\right) \tag{20}$$

Where, $O(\cdot)$ denotes the order of magnitude estimation, n is the number of layers, M_i is the size of the input feature map at the i-th layer, K_i is the kernel size, and C_{i-1}, C_i are the input and output channel numbers of the i-th layer, respectively.

Computational Complexity (GFLOPs): Indicates the number of floating-point operations required per second during model inference, estimated by:

$$\mathrm{GFLOPs} = \frac{1}{10^9}\sum_{i=1}^{n}\left(2K_i^2 C_{i-1} C_i M_i^2\right) \tag{21}$$

Model Size (in MB): Measures the actual storage requirement of the model, typically proportional to the number of parameters. It is calculated as:

$$\text{Model Size(MB)} = \frac{\mathrm{Params}\cdot b}{8\cdot 10^6} \tag{22}$$

where b is the bit-width of each weight parameter, typically 32 bits in standard float precision.

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3.4. Comparative Experiment on the ASF-P Module

To validate the effectiveness of the proposed ASF-P module, we conducted a comparative experiment against the original ASF module. Both modules were individually integrated into the baseline YOLOv8n and trained under identical conditions on the DeTFLy dataset (Section 3.2). The results are summarized in Table 1.

Table 1. Performance and efficiency comparison between ASF and ASF-P modules.

Models	P (%)	R (%)	mAP@0.5 (%)	mAP@0.5:0.95(%)	Parameters (M)	FLOPs (G)	Model size (MB)
YOLOV8n+ASF-P	**95.3**	**83.7**	**90.0**	**56.1**	**2.49**	12.0	**5.3**
YOLOV8n+ASF	95.0	70.8	80.3	47.5	3.05	**8.6**	6.3

The results clearly demonstrate the superiority of ASF-P-enhanced model. As shown in Table 1, it achieves superior precision (95.3%), recall (83.7%), mAP@0.5 (90.0%), and mAP@0.5:0.95 (56.1%). This corresponds to absolute gains of +0.3%, +12.9%, +9.7%, and +8.6%, respectively, over the ASF-based model. The substantial improvement, especially in recall and mAP, indicates that the ASF-P module more effectively handles small UAV targets and complex backgrounds through its enhanced multi-scale feature fusion.

In terms of efficiency, ASF-P also maintains advantages. It reduces the parameter count from 3.05M to 2.49M and model size from 6.3MB to 5.3MB. Although this comes with an increased computational cost (GFLOPs rise from 8.6 to 12.0), the significant gains in accuracy justify this trade-off, and the overall complexity remains within a practical range for deployment.

3.5. Comparative Experiment on the AB-CGLU Module

To evaluate the effectiveness of the proposed AB-CGLU module, we conducted comparative experiments by integrating it and the original AB individually into the YOLOv8n backbone. All models were trained and tested on the DeTFly dataset under identical conditions. The results are presented in Table 2.

Table 2. Performance and efficiency comparison between AB and AB-CGLU modules.

Models	P (%)	R (%)	mAP@0.5 (%)	mAP@0.5:0.95(%)	Parameters (M)	FLOPs (G)	Model size (MB)
YOLOV8n+AB-CGLU	**95.9**	**69.3**	**80.8**	**48.1**	**2.58**	**7.0**	**5.6**
YOLOV8n+AB	94.8	68.2	80.2	47.6	2.89	7.7	6.2

The AB-CGLU-enhanced model shows consistent improvements over the AB baseline. As detailed in Table 2, it achieves higher precision (95.9%), recall (69.3%), mAP@0.5 (80.8%), and mAP@0.5:0.95 (48.1%), corresponding to gains of +1.1%, +1.1%, +0.6%, and +0.5%, respectively. These results confirm that the AB-CGLU module enhances feature representation and detection accuracy, particularly in complex scenarios with small objects.

In terms of efficiency, AB-CGLU demonstrates clear advantages. It reduces the parameter count from 2.89M to 2.58M and the model size from 6.2MB to 5.6MB. Moreover, it lowers the computational cost, with GFLOPs decreasing from 7.7 to 7.0. This combination of improved accuracy and reduced resource demand makes the AB-CGLU module particularly suitable for deployment on resource-constrained edge devices or UAV platforms.

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3.6. Ablation Study

To evaluate the individual and combined contributions of the proposed modules, we conducted an ablation study by incrementally integrating ASF-P, AB-CGLU, and ADown into the YOLOv8n baseline. All experiments used identical hardware, software, dataset, and training configurations to ensure consistency. The results are summarized in Table 3, where "√" and "×" denote the inclusion or exclusion of a module, respectively. Arrows (↑/↓) indicate the desirable direction for each metric.

Table 3. Results of the ablation study.

YOLOv8n	ASF-P	AB-CGLU	ADown	P (%) ↑	R (%) ↑	mAP@0.5 (%) ↑	mAP@0.5:0.95 (%)↑	Parameters (M) ↓	FLOPs (G) ↓	Model size (MB) ↓
√	×	×	×	95.9	70.0	79.0	47.3	3.01	8.2	6.2
√	√	×	×	95.3	83.7	90.0	56.1	2.49	12.0	5.3
√	×	√	×	95.9	69.3	80.8	48.1	2.58	7.0	5.6
√	×	×	√	95.4	70.6	79.9	46.5	2.60	7.5	5.4
√	√	×	√	97.0	**84.7**	91.5	57.0	2.17	10.7	4.9
√	√	√	×	97.0	84.6	91.1	56.6	2.36	10.9	5.1
√	×	√	√	92.3	70.4	81.1	48.1	2.17	**6.3**	4.8
√	√	√	√	**98.1**	84.4	**91.6**	**57.1**	**2.03**	9.6	**4.7**

The results demonstrate that each module contributes distinctively to improving accuracy or efficiency:

- **ASF-P Module:** Its integration boosted mAP@0.5 to 90.0%—an 11.0% absolute gain over the baseline—highlighting its strength in multi-scale feature fusion. It also reduced parameters and model size, though at increased computational cost (12.0 GFLOPs).
- **AB-CGLU Module:** This module improved precision while maintaining a compact form, achieving 80.8% mAP@0.5 with only 7.0 GFLOPs, demonstrating effective context aggregation.
- **ADown Module:** It primarily enhanced efficiency, notably reducing parameters, GFLOPs, and model size, with a modest accuracy gain to 79.9% mAP@0.5, validating its adaptive downsampling design.

The combinations reveal clear synergistic effects:

- **ASF-P + AB-CGLU:** Achieved high accuracy (91.1% mAP@0.5, 84.6% recall) while reducing GFLOPs by 10.8% compared to using ASF-P alone.
- **ASF-P + ADown:** Reached 91.5% mAP@0.5 with further reductions in model size and parameters, striking an excellent accuracy-efficiency balance.
- **AB-CGLU + ADown:** This pairing yielded the most lightweight profile (2.17M params, 6.3 GFLOPs, 4.8MB), ideal for resource-constrained settings, while still improving mAP@0.5 to 81.1%.

Full Integration (Ours): The complete model, integrating all three modules, achieves the highest accuracy (98.1% precision, 91.6% mAP@0.5) and the most compact size (2.03M parameters, 4.7MB) among all configurations. While its computational cost (9.6 GFLOPs) is not the lowest, it represents the optimal trade-off, delivering superior detection performance with high deploy ability for A2A UAV tasks.

3.7. Comparative Experiment with Baseline Model

3.7.1 Quantitative Comparison

We comprehensively evaluate our proposed model against the baseline YOLOv8n on the DeTFly dataset, comparing both detection accuracy and model efficiency, the results summarized in Table 4.

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Table 4. Quantitative comparison between YOLOv8n and the proposed model.

Models	P (%)	R (%)	mAP@0.5 (%)	mAP@0.5:0.95(%)	Parameters(M)	FLOPs (G)	Model size (MB)
YOLOV8n	95.9	70.0	79.0	46.5	3.01	**8.2**	6.2
Ours	**98.1**	**84.4**	**91.6**	**57.1**	**2.03**	9.6	**4.7**

As shown in Table 4, our model achieves superior accuracy across all metrics: 98.1% precision, 84.4% recall, and 91.6% mAP@0.5. These represent substantial absolute gains of +2.2%, +14.4%, and +12.6%, respectively. The stricter mAP@0.5:0.95 also improves markedly from 46.5% to 57.1%, underscoring significantly enhanced localization robustness.

In terms of efficiency, our model is more compact, with parameters reduced from 3.01M to 2.03M and model size from 6.2MB to 4.7MB. This gain in lightweightness comes with a moderate increase in computational cost (GFLOPs from 8.2 to 9.6), a well-justified trade-off for the obtained accuracy improvements, keeping the model within a practical range for deployment.

3.7.2 Precision-Recall Curve Analysis

The detection performance is further compared via the Precision-Recall (P-R) curves of YOLOv8n and our proposed model in Fig. 10. A critical distinction emerges in the mid-to-high recall region (Recall > 0.6), which is most relevant for practical deployment requiring high detection rates. Here, our model maintains stable, high precision, whereas the precision of YOLOv8n undergoes a steep decline. This demonstrates that our model achieves a more robust and favorable precision-recall trade-off, a key attribute for reliably detecting small UAVs amidst complex backgrounds.

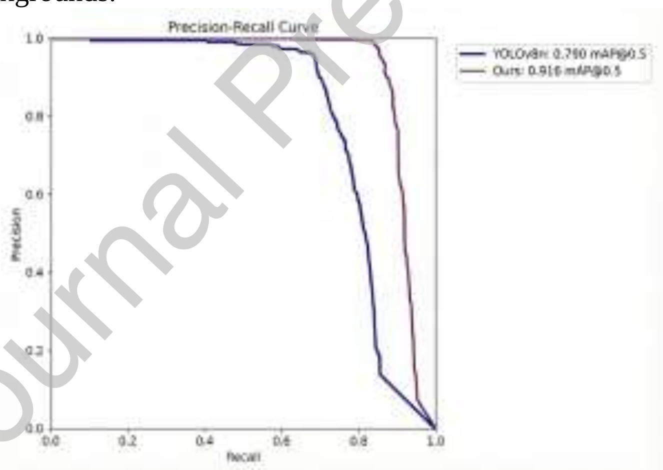

Fig 10. Comparative Precision-Recall (P-R) curves.

3.7.3 Receptive Field Analysis

To probe the internal dynamics of our architectural improvements, we analyzed the gradient flow and receptive fields using 50 randomly sampled images from the DeTFLy validation set. For a clear macroscopic view, raw gradient magnitudes were processed with a moving-average window (20,480 pixels), a $\log_{10}(x+1)$ transform, and min-max normalization to [0, 1], enabling a scale-invariant comparison.

As shown in the gradient maps (Fig. 11, left), YOLOv8n exhibits a dispersed and irregular distribution, whereas our model produces a sharply concentrated pattern. This indicates that during backpropagation, our model's parameters are updated more cohesively around semantically salient regions, promoting stable and efficient learning.

This contrast in gradient flow directly explains the receptive field behaviors (Fig. 11, right). The concentrated gradients in our model yield a focused, target-centric activation that suppresses background interference. In contrast, YOLOv8n's dispersed gradients correspond to a broader, noisier receptive field that is more susceptible to background distractions, underscoring the role of our architectural enhancements in achieving superior spatial selectivity.

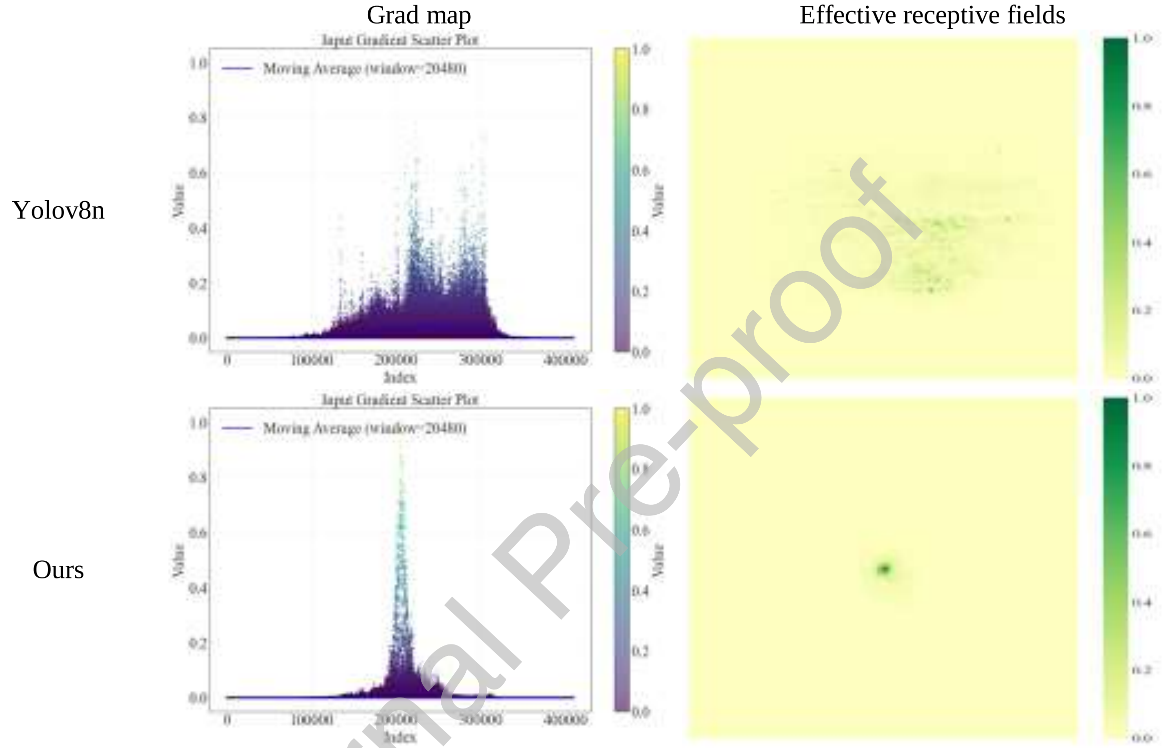

Fig 11. Visualization of grad map and effective receptive fields. In the gradient map, the horizontal axis shows the position of the data points in the index, and the vertical axis shows the gradient value at each position.

3.7.4 Visualization Comparison

For a fair and informative qualitative analysis, a set of representative test images was selected based on the following objective criteria: (1) All images are from the independent test set; (2) They collectively cover typical and challenging scenarios in our A2A dataset, including small targets, complex backgrounds (Field, Sky, Mountain, and Urban), and varying lighting conditions, including low-light and bright-light situations; (3) The model's performance on each selected image is consistent with its average performance on the entire test set. Each figure (Figs. 12, 13, 16, and 19) independently selects images according to these criteria for visual comparison.

Fig 12 presents a direct comparison of detection results on these representative images. It is evident that across diverse backgrounds, our model achieves more accurate bounding box localization, higher prediction confidence, and superior background suppression compared to YOLOv8n.

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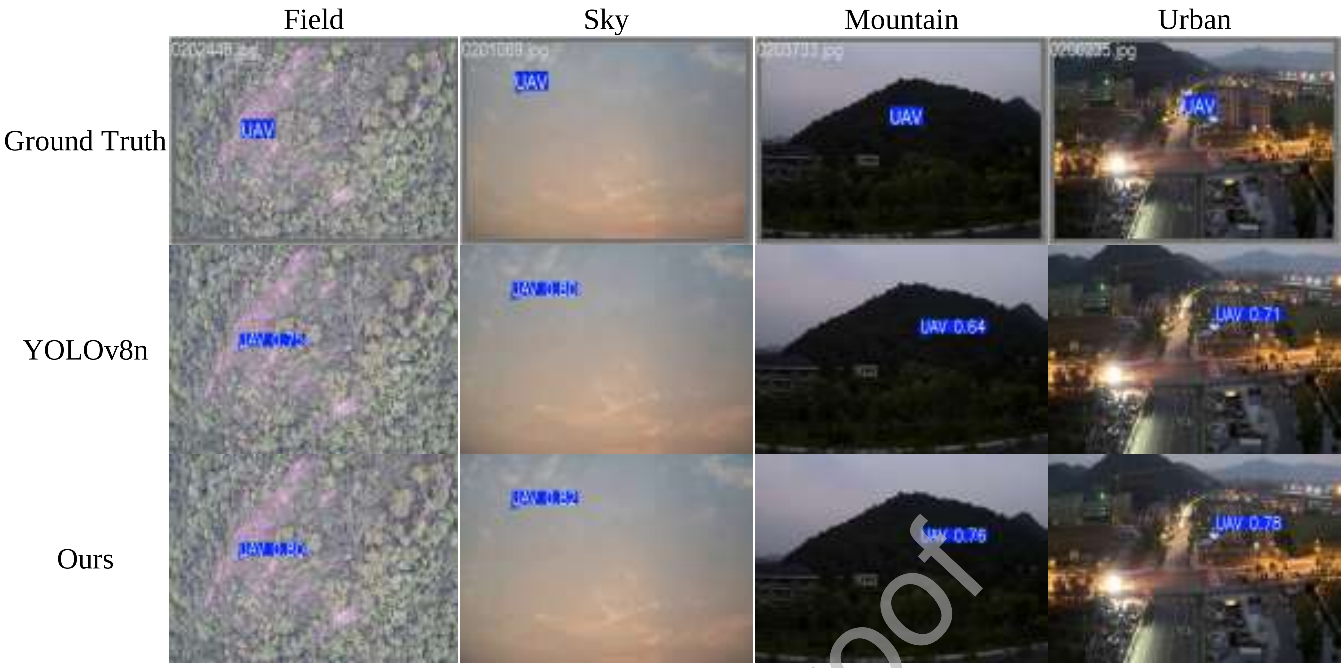

Fig 12. Qualitative detection comparison on representative test images. From top to bottom: Ground truth; Detection results of YOLOv8n; Detection results of our model. Columns represent different background types: Field, Sky, Mountain, Urban.

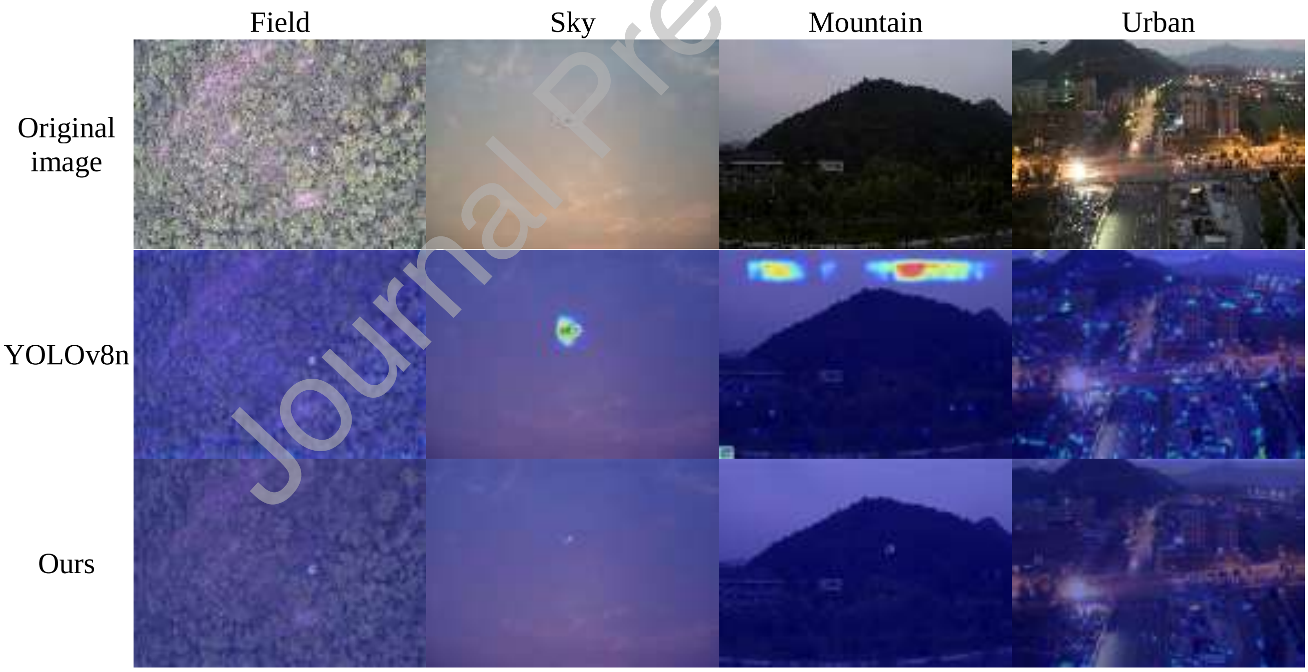

Fig 13. Comparison of activation heatmaps on representative images. From top to bottom: Original input image; Activation heatmap of YOLOv8n; Activation heatmap of our model.

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Further insight is provided by the activation heatmaps in Fig 13. Compared to YOLOv8n, our model exhibits sharper and more focused activation on the actual UAV regions, effectively suppressing irrelevant background noise. This visual evidence confirms the enhanced feature discriminativity and robustness of our proposed model in complex real-world scenarios.

3.8. Comparative Experiments with State-of-the-Art YOLO Models

We conducted comprehensive comparisons with state-of-the-art YOLO models to rigorously evaluate the performance and generalizability of our proposed framework. To account for the diversity in model complexity and design, our comparisons are organized into two categories: (1) Lightweight (nano) series: YOLOv8n through YOLOv12n. (2) Small series: YOLOv8s through YOLOv12s. For each category, we provide a multi-faceted evaluation, including: Quantitative metric comparison, Radar-chart visualization of balanced performance, Precision-Recall (P-R) curve analysis, and Qualitative visual comparison of detection results.

3.8.1 Comparison with Nano-Series YOLO Models

We comprehensively evaluate our model against leading YOLO Nano-series models (YOLOv9t, v10n, v11n, v12n) under identical settings on the DeTFLy dataset.

Table 5 summarizes the quantitative comparison. As shown in Table 5, our model significantly outperforms all counterparts across key metrics. It achieves +7.5% to +13.5% higher mAP@0.5 and +3.6% to +10.2% higher mAP@0.5:0.95. Notably, it attains a precision of 98.1% and recall of 84.4% while maintaining the smallest size (2.03M parameters, 4.7MB), demonstrating an exceptional accuracy-efficiency trade-off.

Table 5. Performance comparison with YOLO Nano-series models.

Models	P (%) ↑	R (%) ↑	mAP@0.5 (%)↑	mAP@0.5:0.95 (%) ↑	Parameters (M) ↓	FLOPs (G) ↓	Model size (MB) ↓
YOLOv9t	94.8	72.0	83.0	51.6	2.66	11.0	6.1
YOLOv10n	89.0	69.4	78.1	47.1	2.71	8.4	5.8
YOLO11n	95.4	75.5	84.1	53.5	2.59	6.4	5.5
YOLOv12n	94.5	68.3	78.1	46.9	2.52	**6.0**	5.4
Ours	**98.1**	**84.4**	**91.6**	**57.1**	**2.03**	9.6	**4.7**

The radar-chart (Fig 14) and Precision-Recall curves (Fig 15) provide intuitive comparisons.

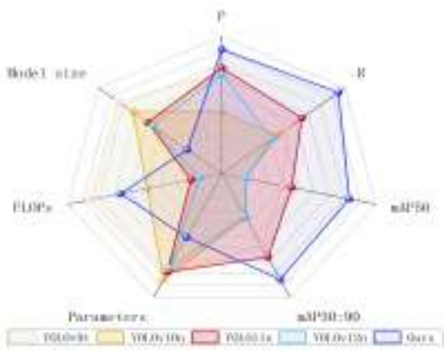


Fig 14. Comparative radar chart for the YOLO Nano-series models and our model.

In Fig. 14, our model encloses the largest area in accuracy metrics (Precision, Recall, mAPs) while also leading in model size and parameters, confirming an efficient lightweight design. Its slightly higher FLOPs than YOLO11n/YOLOv12n represent a justified trade-off for substantially enhanced performance. Fig. 15 shows our model's P-R curve is smoother and closer to the upper-right corner, maintaining higher precision across recall levels. This indicates superior stability, whereas other models exhibit greater fluctuation and precision drops at higher recall.

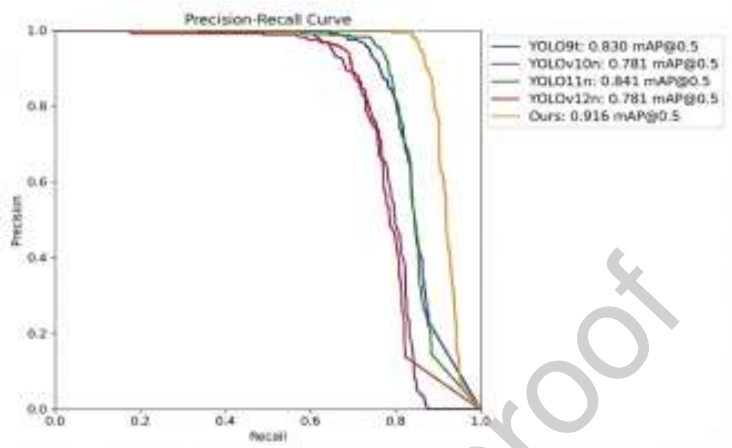


Fig 15. Comparative P-R curves of the YOLO Nano-series models and our model.

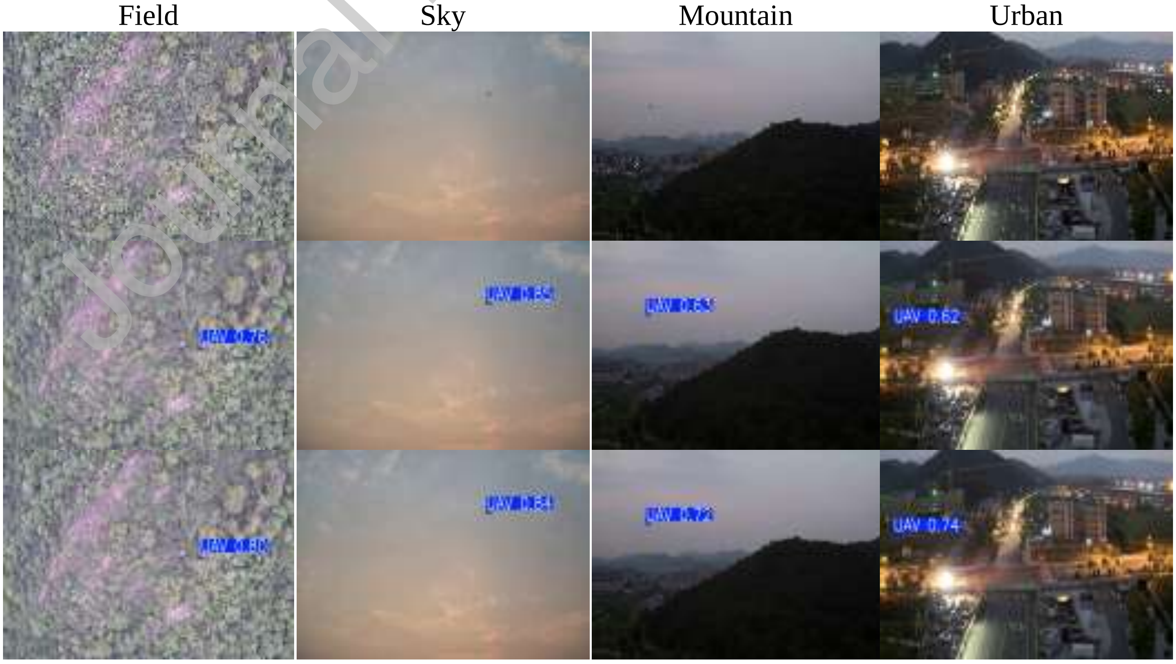
Fig. 16 visually compares the detection results. Our model demonstrates more stable detection across multi-scale UAVs, with more accurate bounding boxes and higher confidence scores, showcasing better robustness and generalization in diverse backgrounds.

Field | Sky | Mountain | Urban

Original image

YOLOv8n

YOLOv9t



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YOLOv10n

YOLO11n

YOLOv12n

Ours

Fig 16. Visual detection results comparison of YOLO Nano series and our model.

3.8.2 Comparison with Small-Series YOLO Models

We further compare our model against more computationally intensive YOLO Small-series models (YOLOv8s-YOLOv12s) to evaluate its performance-efficiency trade-off. These models, designed for precision-critical scenarios, serve as strong benchmarks for accuracy. All experiments were conducted on the DeTFLy dataset under identical settings.

Table 6. Performance comparison with YOLO small series

Models	P (%) ↑	R (%) ↑	mAP@0.5 (%) ↑	mAP@0.5:0.95(%) ↑	Parameters (M) ↓	FLOPs (G) ↓	Model size (MB) ↓
YOLOv8s	95.8	75.2	84.5	51.2	11.13	28.4	21.5
YOLOv9s	95.2	75.4	84.8	54.2	9.74	39.6	20.3
YOLOv10s	93.6	71.6	81.9	50.2	8.07	24.8	16.5
YOLO11s	96.2	80.9	87	56.0	9.43	21.5	19.2
YOLOv12s	94.5	75.8	84.0	50.7	9.10	19.6	18.6
Ours	**98.1**	**84.4**	**91.6**	**57.1**	**2.03**	**9.6**	**4.7**

As shown in Table 6, our model achieves the highest scores across all key accuracy metrics: 98.1% precision, 84.4% recall, 91.6% mAP@0.5, and 57.1% mAP@0.5:0.95, surpassing all five Small-series variants. This

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performance lead is achieved with exceptional efficiency. In item of efficiency, our model requires only 2.03M parameters, 9.6 GFLOPs, and a 4.7MB size. In contrast, the best-performing competitor, YOLO11s, uses 9.43M parameters, 21.5 GFLOPs, and 19.2MB, over 4× our model's footprint. This demonstrates a superior balance between accuracy and lightweight design.

The radar chart (Fig 17) and P-R curves (Fig 18) provide intuitive confirmation. Fig 17 shows our model's balanced and compact profile across all metrics, leading in accuracy while remaining the smallest. Fig. 18 shows its P-R curve encloses the largest area, maintaining higher precision at all recall levels. This stability is crucial for detecting low-confidence small targets, whereas other models show greater performance drops as recall increases.

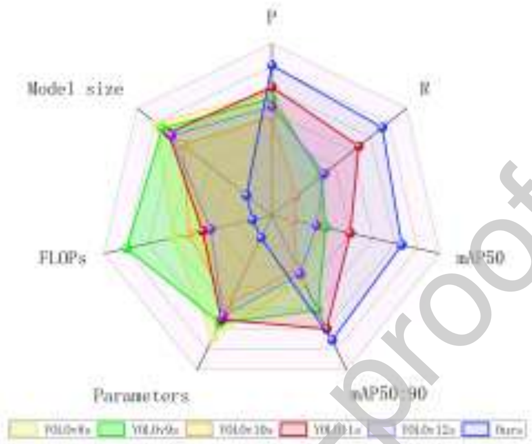

Fig 17. Comparative radar chart for YOLO small series and our model.

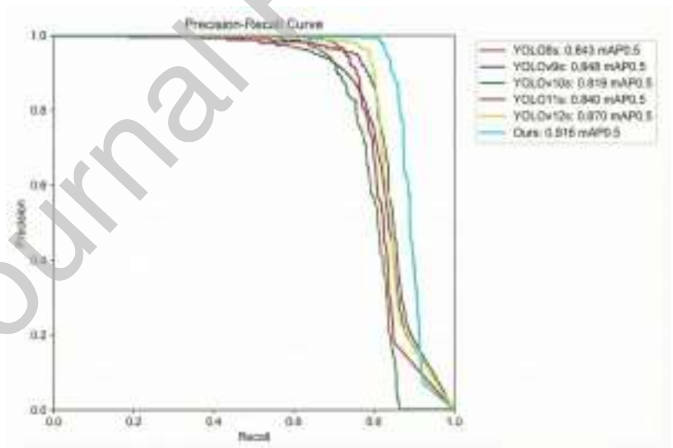

Fig 18. Comparative P-R curves for YOLO small series and our model.

Visual comparisons confirm our model's practical advantage. As highlighted in Fig. 19, baseline models (e.g., YOLOv8s and YOLOv9s in the first column) exhibit missed detections, and all YOLOv8s-YOLOv12s models fail to detect the target in the challenging urban scene (last column). In stark contrast, our model

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consistently provides more accurate localization and higher confidence across all scenes, especially for UAVs in cluttered backgrounds, with low contrast, or under partial occlusion.

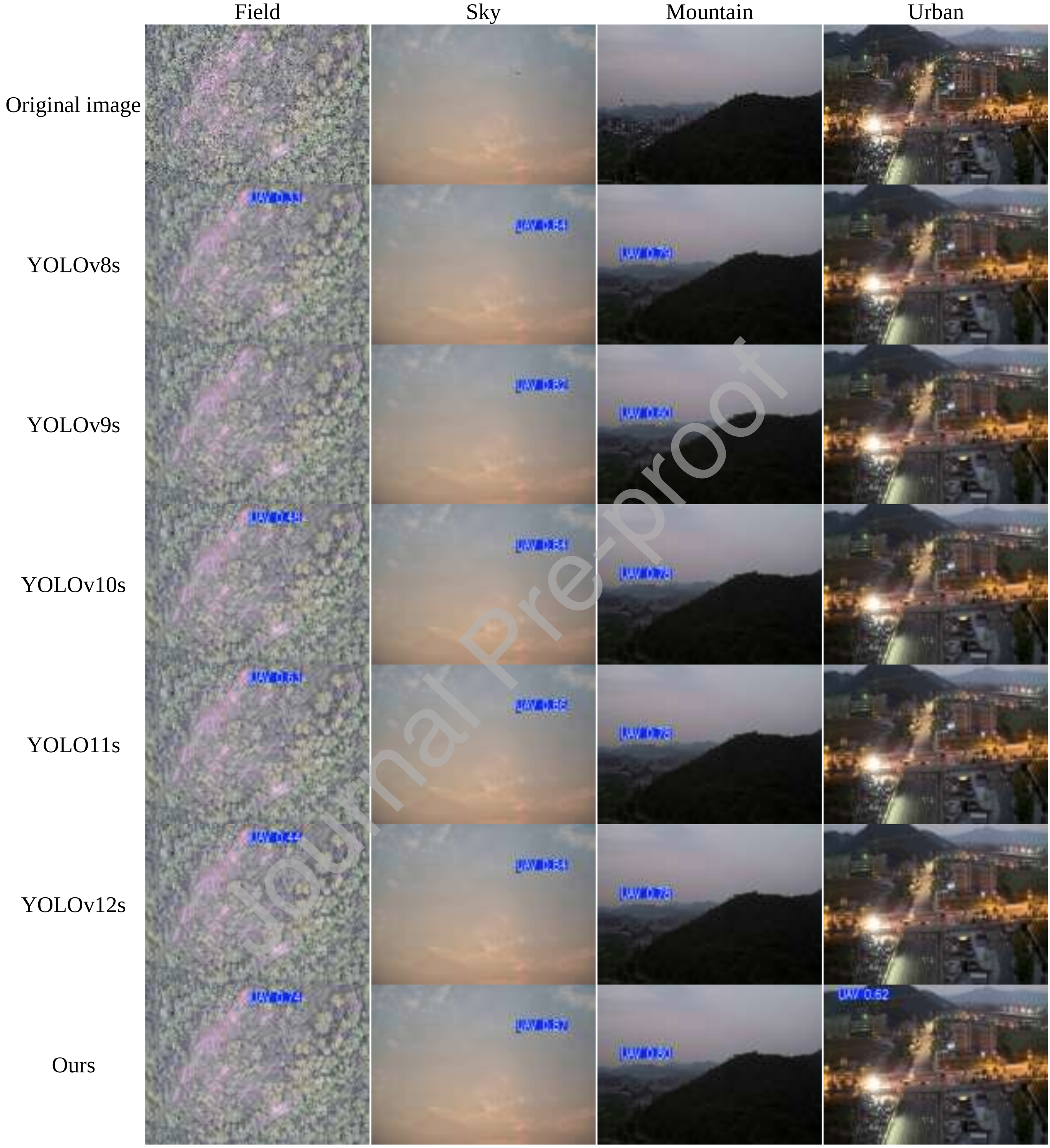

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Fig 19. Visual detection result of YOLO small series and our model.

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3.9. Cross-Dataset Generalization Validation

To directly validate the generalization of our model, we conduct rigorous zero-shot evaluation on two challenging, unseen benchmarks: the LRDDv2 dataset [35] and the DUT Anti-UAV dataset [36]. LRDDv2 features a large variety of cluttered backgrounds (e.g., Multi-target, Occluded, Long-Distance) posing a test of background suppression. It contains a total of 19,885 images, partitioned into 15,900 for training, 1,987 for validation, and 1,998 for testing. DUT Anti-UAV is specifically designed for counter-UAV tasks and contains numerous extremely small, low-resolution, and fast-moving targets under complex scenes. This dataset comprises 10,000 images, with 5,200, 2,600, and 2,200 images allocated for training, validation, and testing, respectively. Critically, our final model (trained solely on DeTFLy) was applied without any fine-tuning. To ensure the reliability of our quantitative results and avoid bias from a single run, we repeat each experiment three times with different random seeds (seed=0, 123, 2024) and report the mean and standard deviation across these runs. All evaluations use the same metrics and settings as the main experiments to ensure a fair comparison with the YOLOv8n baseline.

3.9.1 Quantitative Performance Comparison Across Datasets

The quantitative results are summarized in Table 7 and Table 8. As shown, our proposed model achieves competitive performance on both external datasets.

Table 7. Performance comparison on LRDDv2 dataset

Models	Seed	P (%)	R (%)	mAP@0.5 (%)	mAP@0.5:0.95(%)
YOLOV8n	Seed = 0	95.5	87.1	94.4	67.9
	Seed = 123	94.9	87.0	94.0	67.4
	Seed = 2024	95.7	86.7	94.7	68.1
	Mean ± Std	**95.4±0.3**	**86.9±0.2**	**94.3±0.3**	**67.8±0.3**
Ours	Seed = 0	97.1	94.9	97.5	70.4
	Seed = 123	97.6	95.1	97.7	70.3
	Seed = 2024	96.4	94.2	97.0	69.9
	Mean ± Std	**97.0±0.6**	**94.7±0.5**	**97.4±0.4**	**70.2±0.3**

Table 8. Performance comparison on DUT Anti-UAV dataset

Models	Seed	P (%)	R (%)	mAP@0.5 (%)	mAP@0.5:0.95(%)
YOLOV8n	Seed = 0	92.4	75.3	83.5	53.4
	Seed = 123	91.9	75.2	83.2	53.1
	Seed = 2024	92.2	75.0	83.7	53.6
	Mean ± Std	**92.2±0.3**	**75.2±0.2**	**83.5±0.3**	**53.4±0.3**
Ours	Seed = 0	96.3	83.4	90.0	58.3
	Seed = 123	94.9	83.9	89.6	58.5
	Seed = 2024	95.3	83.8	89.5	58.3
	Mean ± Std	**95.5±0.7**	**83.7±0.3**	**89.7±0.3**	**58.4±0.1**

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The results lead to several key observations:

- **Consistent and Significant Improvements:** In terms of mean performance over three independent training runs. Our model consistently outperforms the baseline YOLOv8n across all metrics on both datasets. Notably, the improvement is more pronounced on the more challenging DUT Anti-UAV dataset, where our model elevates the mean mAP@0.5 by 6.2 percentage points (from 83.5% to 89.7%) and recall (R) by 8.5 percentage points (from 75.2% to 83.7%). This demonstrates that our architectural enhancements provide a greater advantage in difficult scenarios with extremely small and fast-moving targets.
- **Enhanced Robustness to Background Clutter:** On the LRDDv2 dataset, which tests generalization to diverse cluttered backgrounds, our model achieves a 7.8 percentage point increase in mean recall (from 86.9% to 94.7%) and a 3.1 percentage point increase in mean mAP@0.5 (from 94.3% to 97.4%). The simultaneous significant rise in both mean precision (P) and mean recall indicates that our model not only detects more true positives but also maintains high confidence in its predictions, effectively suppressing false alarms in complex environments.
- **Superior Localization Accuracy:** The gains in the stricter mAP@0.5:0.95 metric further confirm the improved bounding box regression quality. On DUT Anti-UAV, our model improves the mean mAP@0.5:0.95 from 53.4% to 58.4%, and on LRDDv2, from 67.8% to 70.2%. These consistent improvements across IoU thresholds indicate that the predicted boxes are more precisely aligned with the ground truth.
- **Training Stability across Random Seeds:** To assess whether the reported improvements are robust to random initialization and data shuffling, each experiment was repeated three times with different random seeds (0, 123, and 2024). The individual results for all seeds are reported in Tables 7–8, alongside the Mean ± Std summaries. Our model exhibits minimal performance fluctuation across seeds on both datasets. For instance, on DUT Anti-UAV, the mAP@0.5 values across the three seeds are 90.0%, 89.6%, and 89.5%, yielding a standard deviation of only ±0.3%. On LRDDv2, the corresponding mAP@0.5 values are 97.5%, 97.7%, and 97.0%, with a standard deviation of ±0.4%. Critically, even the worst-performing seed of our model consistently surpasses the best-performing seed of YOLOv8n by a clear margin. On DUT Anti-UAV, the lowest mAP@0.5 of our model (89.5%) exceeds the highest of YOLOv8n (83.7%) by 5.8 percentage points. Similarly, on LRDDv2, our lowest mAP@0.5 (97.0%) exceeds YOLOv8n's highest (94.7%) by 2.3 percentage points. The same trend holds across all other metrics and both datasets. These observations confirm that the performance gains are not artifacts of a particular favourable weight initialization but reflect a genuine and stable improvement inherent to the proposed architectural modifications.

These quantitative results provide strong evidence that our framework generalizes effectively to unseen domains and retains, even amplifies, its performance advantage as task difficulty increases.

3.9.2 Qualitative Analysis Across Datasets

To intuitively assess the model's robustness under practical challenges, we visualize detection results on representative samples from both new datasets. Fig 20 and Fig 21 showcase outputs across several common yet difficult scenarios.

The visual comparisons in Fig 20 and Fig 21 reveal two key differences in model behavior. First, in multi-target scenarios on LRDDv2 (Fig 20, column 1), YOLOv8n exhibits missed detections, whereas our method consistently identifies and localizes all viable targets. Second, for those objects that are detected by both models in challenging conditions (e.g., Complex background, Fog/Haze), the confidence scores predicted by YOLOv8n are noticeably lower than those of our model.

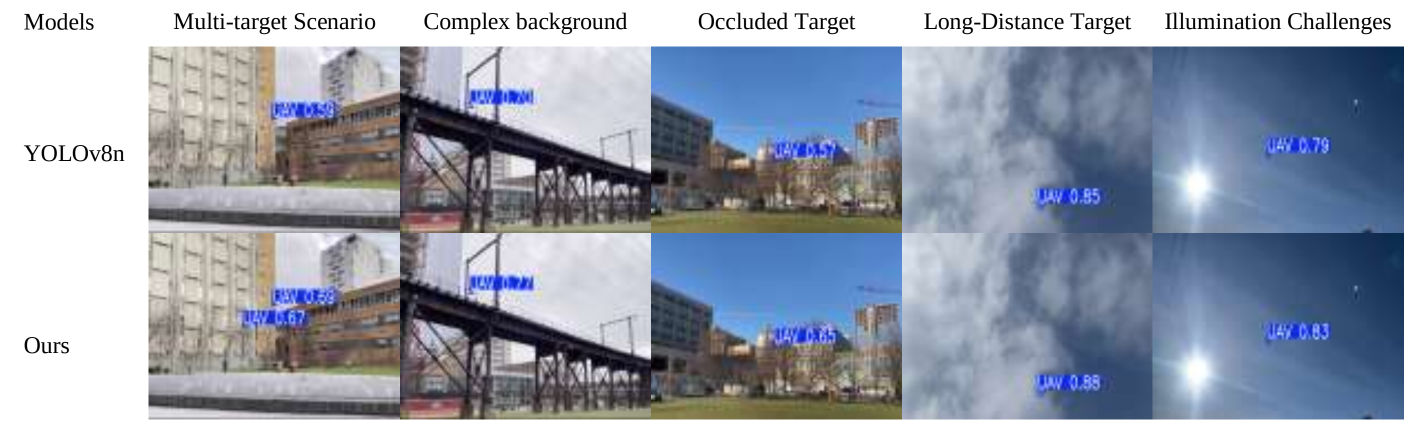

Fig 20. Visual detection comparison on the LRDDv2 dataset.

Fig 21. Visual detection comparison on the DUT Anti-UAV dataset under extreme environmental conditions.

These visual comparisons validate our design: the reduced missed detections stem from the enhanced feature sensitivity of the AB-CGLU and ASF-P modules, while the consistently higher confidence scores under domain shift reflect the more discriminative features learned via our refined attention mechanisms.

4. Discussion

The proposed model demonstrates outstanding performance in A2A UAV detection, offering an efficient and practical solution for small object detection on resource-constrained platforms. Extensive experiments confirm that it achieves an excellent balance between detection accuracy and computational efficiency.

4.1. Overall Performance and Lightweight Design

On the primary DeTFLy benchmark, the proposed model achieves an excellent balance between accuracy and efficiency. Compared to YOLOv8n, it raises mAP@0.5 by 12.6% (to 91.6%) and mAP@0.5:0.95 by 10.6% (to 57.1%), while reducing parameters by 32.6% (to 2.03M) and model size by 24.2% (to 4.7MB). This makes it highly suitable for embedded deployment on UAV platforms.

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4.2. Contributions of Individual Modules

Ablation studies further demonstrate the individual contributions and synergy of the proposed modules. The ASF-P module significantly improves multi-scale representation and small-object detection capability, increasing mAP@0.5 to 90.0% when used alone. The AB-CGLU module enhances key feature selection via spatial–channel interaction and gating mechanisms, reducing parameters to 2.58M and FLOPs to 7.0G. The ADown module optimizes the downsampling path while maintaining accuracy, achieving 2.60M parameters and 7.5G FLOPs. The integration of all three modules yields the complete model's performance: 98.1% precision, 84.4% recall, 91.6% mAP@0.5, and 57.1% mAP@0.5:0.95.

4.3. Comparison with State-of-the-Art Models

Compared with mainstream lightweight and high-performance detectors such as YOLOv9t, YOLOv12n, and YOLOv11s, our model consistently delivers superior performance. For instance, relative to YOLOv11s, it improves mAP@0.5 by 4.6 percentage points while reducing parameters by 78.5%, FLOPs by 55.3%, and model size by 75.5%, showcasing an excellent performance-efficiency trade-off.

4.4. Generalization Capability

Beyond the primary experiments, the cross-dataset validation (Section 3.9) provides strong evidence for the model's generalization. In a zero-shot transfer setting, our model trained solely on DeTFLy maintains superior performance on both the LRDDv2 (cluttered backgrounds) and DUT Anti-UAV (extremely small, fast-moving targets) datasets. As summarized in Tables 7–8 (mean ± std over three independent runs with different random seeds), the improvements are both substantial and stable. Notably, on the more challenging DUT Anti-UAV, our model elevates mean mAP@0.5 by 6.2% and the mean recall by 8.5% compared to YOLOv8n. The standard deviations across seeds are minimal ($\leq$ 0.3% for mAP@0.5), and even the lowest single-seed result of our model outperforms the highest YOLOv8n result. On LRDDv2, the mean recall improves by 7.8 percentage points (86.9% $\rightarrow$ 94.7%) and the mean mAP@0.5 by 3.1 percentage points (94.3% $\rightarrow$ 97.4%), again with tight variance. This demonstrates that the architectural enhancements confer robust feature representations that generalize effectively to unseen domains and increased difficulty, and that the gains are statistically reliable rather than dependent on a particular training run.

4.5. Limitations and Future Work

The current design prioritizes a balance between accuracy and lightweight structure. While effective, this results in a computational cost (9.6 GFLOPs) higher than some extreme-lightweight variants (e.g., YOLOv8n's 6.2G). This may impact inference speed on severely power-constrained platforms. Future work will explore more efficient attention mechanisms, neural architecture search, and model compression techniques to further optimize deployment efficiency without compromising the demonstrated robustness and accuracy.

5. Conclusion

This study presents an efficient and lightweight YOLO-based framework tailored for A2A UAV detection, effectively addressing the challenges of small targets in complex backgrounds. The proposed architecture integrates three key modules: the ASF-P for enhanced multi-scale and small-object feature fusion, the AB-

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CGLU for refined contextual feature interaction and selection, and the ADown for efficient downsampling. Together, they significantly boost detection capability while maintaining a compact model structure.

Extensive experiments on the primary dataset demonstrate that our model achieves a precision of 98.1%, recall of 84.4%, mAP@0.5 of 91.6%, and mAP@0.5:0.95 of 57.1%, with only 2.03M parameters and a 4.7MB size. Compared to YOLOv8n, this represents an improvement of 12.6% in mAP@0.5 alongside a 32.6% reduction in parameters, showcasing an exceptional accuracy-efficiency trade-off. Critically, rigorous cross-dataset over three independent training runs with different random seeds validation confirms the model's strong generalization capability. Under a strict zero-shot transfer protocol, our model maintains superior performance on the unseen LRDDv2 and DUT Anti-UAV benchmarks, with particularly pronounced gains on the latter's challenging small and fast-moving targets. The low standard deviations observed across seeds (e.g., mAP@0.5: 97.4 ± 0.4% on LRDDv2, 89.7 ± 0.3% on DUT Anti-UAV) verify that the proposed enhancements learn robust and transferable feature representations.

Overall, our work provides a practical, high-performance solution for A2A UAV detection that is readily deployable on resource-constrained platforms. It also offers a coherent architectural strategy for lightweight YOLO optimization, presenting valuable insights for future research in efficient aerial perception.

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Declaration of interests

☐The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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